Agentic Development Reference Lab

Individual Software Engineering Deliverable 1

LIBRARY MANAGEMENT REST API

A Java 17-compatible Spring Boot 3 reference application developed through eight progressive stages with test-first, specification-driven, review, quality, and deployment evidence.

8	21/21	82.03%	71.88%
progressive stages	final tests passing	instruction coverage	branch coverage

Student account	Branch	Repository
Yorgo756	yorgo/reference-lab	github.com/Yorgo756/genie-logiciel-reference-lab

Evidence note

This report is based on committed source, test reports, raw command logs, generated quality reports, and live Docker HTTP checks. No screenshots were fabricated. Where a screenshot was not captured, a genuine terminal excerpt is used and labeled.

Official-handout note

The official handout was unavailable. Requirements, routes, quality thresholds, and function-point classifications are therefore explicitly provisional and can be reconciled later.

1. Method, scope, and environment

The work used a repeatable agentic loop: inspect the current state, plan a narrow increment, implement it, run executable verification, review failures and design choices, correct defects, and commit meaningful evidence.

Element	Verified configuration	Evidence role
Application	Spring Boot 3.5.16; Java release 17	REST API and domain behavior
Data	Spring Data JPA; H2 in-memory	Repositories and reproducible seeded state
Build	Maven Wrapper 3.10.0	Portable compile, test, package, analysis
Tests	JUnit 5, MockMvc, Mockito	Unit, integration, HTTP, error workflows
API docs	springdoc OpenAPI / Swagger UI	Machine and browser-readable contract
Deployment	Multi-stage Dockerfile; Temurin 17.0.21	Verified isolated runtime
Quality	PMD 3.28.0; SpotBugs 4.10.5.0; JaCoCo 0.8.16	Static findings and measured coverage

Provisional scope decisions

- Books, members, borrowings, and reservations are separate persisted resources with generated numeric identifiers.
- Loans last 14 calendar days; overdue charges are EUR 0.50 per whole late day and stop at return.
- Reservation ownership is the member identifier in the path; authentication and authorization are outside the provisional scope.
- H2 state is recreated at startup with five books and three active members.

Environment issue and response

Spring Initializr exposed Boot 4 metadata and returned HTTP 400 for the requested 3.5.16 archive. Maven Central was checked, and the Boot 3.5.16 scaffold was created explicitly with the official Maven Wrapper. The host runs Oracle JDK 22.0.2 while Maven compiles for release 17.

Agent interaction evidence

```
Student: Continue through Labs 2-8; use real tests, reviews, commits,
         Docker verification, estimates, and a Moodle PDF.
Agent:   Preserved RED/GREEN evidence, corrected observed failures,
         and reported provisional assumptions and remaining limits.
```

2. Eight-stage workflow: Labs 1-4

Lab	Implemented and reviewed	Verification and commits
1 - Scaffold	Book, Member, Borrowing entities; repositories, services, controllers; CRUD; H2; five books and three members.	Build, automated tests, and HTTP checks. 7748938, d500522
2 - Agile/TDD	Sprint backlog and four stories: borrow 14 days, return/availability, active loans, EUR 0.50 late fee. Duplicate and invalid states rejected.	RED: 4/4 failed with 404. GREEN: 12 passed. 4f0532e, 801f5bd
3 - Specification	Acceptance criteria committed before production code. Reservations require unavailable books and active members; duplicates blocked; owner list/cancel.	RED: 3/3 failed with 404. GREEN: 15 passed. 442514a, a0abf14, 8192839
4 - Design/deploy	DTO boundary, mappers, service separation, domain exceptions, global ApiError, OpenAPI/Swagger, multi-stage Dockerfile.	Container and HTTP verified. 77d7781

Lab 2 RED to GREEN excerpt

```
RED   ./mvnw -Dtest=BorrowingWorkflowIntegrationTest test
      Tests run: 4, Failures: 4 - expected 201, received 404
GREEN ./mvnw test
      Tests run: 12, Failures: 0, Errors: 0, Skipped: 0
```

Lab 3 specification review

- Create reservation: HTTP 201 only when the book is unavailable and the member is active.
- Missing book/member/reservation: HTTP 404. Available book, inactive member, or duplicate active reservation: HTTP 409.
- Owner-specific list: HTTP 200. Owner cancellation: HTTP 204; repeat or non-owner cancellation is hidden as 404.

Design review decision

The generic Lab 1 borrowing endpoint delegates to the same 14-day service workflow as the member route, preventing a bypass of availability and membership rules. Services throw domain exceptions; one controller advice maps them to a stable timestamp/status/error/message/path body.

3. Labs 5-8: refinement and evidence

Lab	Implemented and reviewed	Verification and commits
5 - Refactoring	1) Extract LoanPolicy. 2) Encapsulate book/loan transitions. 3) Make active-loan endpoint single-purpose.	17 tests passed after each small change. baef2ee, ddb4f9e, 3d1b47e
6 - Quality	Configured PMD, SpotBugs, JaCoCo and GQM. Corrected three MissingSerialVersionUID findings.	PMD 3 to 0; coverage 81.09%/68.75% at stage; 18 SpotBugs warnings disclosed. 9c384f3
7 - Test quality	Added error-shape, validation, missing-resource, and completed-fee workflows. Mutated loan term 14 to 13, then restored it.	Mutation test failed exactly as expected; restored suite 21/21. 1bbb2a1, 365dd7b, d0a288b
8 - Estimation	129 provisional UFP with visible classifications; three productivity scenarios; evidence-based retrospective.	No actual hours or measured productivity invented. 98b78e1

Three refactorings: before and after

Concern	Before	After
Loan policy	Magic 14 and 0.50 inside service	LoanPolicy owns due-date and fee rules
State changes	Service toggled fields directly	Book and Borrowing name their transitions
Active query	Boolean parameter rejected false	Single-purpose active-list operation

Lab 7 deliberate-defect proof

```
Temporary change: LOAN_DAYS = 13 (not committed)
Tests run: 1, Failures: 1
JSON path "$.dueDate" expected:<2026-11-24> but was:<2026-11-23>
Restored: LOAN_DAYS = 14
Tests run: 21, Failures: 0, Errors: 0, Skipped: 0
```

Integrity decision

The broken implementation was never committed. The unedited failing log and the full restored-suite log were committed, while Git diff confirmed LoanPolicy matched the correct baseline.

4. API design and live deployment

The design separates HTTP DTOs and mapping from transactional services and JPA entities. The same domain workflow serves generic and member-scoped borrowing routes, and one exception boundary produces consistent 400, 404, and 409 responses.

HTTP boundary	Service/domain	Persistence/output
Validated request DTOs	Book, member, loan, reservation rules	Four Spring Data repositories
Controllers + mappers	LoanPolicy and state transitions	H2 for development/test
Global ApiError advice	Domain exceptions, transactions	Response DTOs + OpenAPI

Docker problem and correction

The first `eclipse-temurin:17-jdk-alpine` build lacked a usable ARM64 manifest. The Dockerfile was changed to Jammy JDK/JRE stages. The final image built with manifest-list digest `sha256:1ba0a7d...8ce9fdac` and ran with Temurin 17.0.21.

Final live-container HTTP checks

Request/check	Observed result
GET /api/books	200; five seeded books
GET /v3/api-docs	200; Library Management API; 11 paths
GET /swagger-ui/index.html	200
POST invalid book	400; standard ApiError
POST member 1 borrowing	201; due 2026-11-24; unavailable
POST member 2 reservation	201; book 1 reserved
DELETE owned reservation	204
PUT return	200; available true
PUT second return	409; already returned

Genuine command excerpt

```
books_count=5
openapi_title_and_paths=["Library Management API",11]
borrow_status=201 body={"dueDate":"2026-11-24","available":false}
reservation_status=201 ... cancel_status=204
return_status=200 ... second_return_status=409
```

5. Quality and final verification

21/21	82.03%	71.88%	0
tests passing	instructions	branches	PMD violations

GQM question/metric	Provisional target	Final actual	Assessment
Automated test result	100% pass	21 pass; 0 fail/error/skip	Met
JaCoCo instructions	>= 70%	1,219/1,486 = 82.03%	Met
JaCoCo branches	>= 60%	23/32 = 71.88%	Met
PMD	Report and correct genuine finding	3 initial; 0 after correction	Met
SpotBugs	Report actionable findings	18 priority-2 warnings disclosed	Reviewed
HTTP behavior	201/204/400/404/409 tested	All represented in workflows	Met

Static-analysis review

PMD identified `MissingSerialVersionUID` in three custom exceptions. Adding the fields was behavior-preserving and reduced PMD from three to zero. SpotBugs still reports 4 `EI_EXPOSE_REP` and 14 `EI_EXPOSE_REP2` warnings around mutable JPA entity references and Spring-injected collaborators. They remain visible rather than being suppressed with a broad filter.

Final clean command

```
./mvnw clean verify pmd:pmd spotbugs:spotbugs
Tests run: 21, Failures: 0, Errors: 0, Skipped: 0
BUILD SUCCESS | Total time: 14.152 s
JaCoCo: instructions 82.03%, branches 71.88%
PMD: 0 | SpotBugs: 18 disclosed warnings
```

Known warning

Host JDK 22 reports Mockito's future dynamic-agent compatibility warning. It did not fail compilation or tests and does not alter the Java 17 container runtime; it is retained as a follow-up item.

6. Estimation and retrospective

The estimate is a classroom planning exercise, not a measurement. The API boundary excludes Swagger UI, build tools, tests, and seed data. Average IFPUG weights are used because data-element and record-element thresholds were not specified.

Type	Count	Weight	Points	Basis
EI	13	4	52	Create/update/delete and workflow commands
EO	1	5	5	Calculated late-fee response
EQ	8	4	32	List/get and member-specific queries
ILF	4	10	40	Books, members, borrowings, reservations
EIF	0	7	0	No external interface file
Total UFP			129	No value-adjustment factor

Hypothetical effort scenarios

Assumed productivity	Formula	Hypothetical effort
8 FP/person-month	129 / 8	16.13 person-months
12 FP/person-month	129 / 12	10.75 person-months
16 FP/person-month	129 / 16	8.06 person-months

No measured-hours claim

No start/stop time log or independently measured development-hours record exists. This submission therefore reports no actual hours and no measured productivity. Commits, tests, coverage, and command results are engineering evidence, not substitutes for elapsed time.

Retrospective

- Worked: small verified increments, RED/GREEN history, specification before reservations, and regression after each refactoring.
- Corrected: Initializr 400, reservation 404s before implementation, ARM64 Alpine image failure, and PMD serial-version findings.
- Next: reconcile the official handout; add migrations and a persistent profile; add authentication and concurrency tests; review a narrow framework-aware SpotBugs policy.

7. Evidence map, limitations, and conclusion

The repository preserves one report per stage plus raw logs for the deliberate defect, restored suite, final clean verification, and final Docker HTTP run.

Evidence	Repository location	Purpose
Stage reports	reference-lab/reports/lab1-report.md ... lab8-report.md	Commands, results, decisions, issues, commits
Interaction record	reference-lab/reports/agent-interactions.md	Concise genuine request/action history
RED/GREEN artifacts	docs/sprint-backlog.md; docs/reservation-specification.md	Backlog and acceptance criteria written before code
Quality plan	docs/gqm.md	Questions, metrics, provisional targets
Mutation logs	reports/evidence/lab7-*.txt	Failure and restored-suite output
Final logs	reports/evidence/final-*.txt	Clean build and container HTTP output
Estimation/reflection	docs/function-point-estimate.md; docs/retrospective.md	Assumptions, scenarios, learning

Representative Git history

Commit(s)	Meaningful increment
4f0532e -> 801f5bd	Lab 2 lending RED to GREEN
442514a -> a0abf14 -> 8192839	Lab 3 specification, failing acceptance tests, implementation
baef2ee / ddb4f9e / 3d1b47e	Three independently verified refactorings
9c384f3	Quality tooling, metrics, and PMD correction
1bbb2a1 / 365dd7b / d0a288b	Expanded tests, mutation evidence, recovery report
98b78e1 / 38c6417	Estimation, reflection, and final verification evidence

Remaining limitations

- Official-handout alignment is unverified; all such assumptions are labeled provisional.
- H2 data is ephemeral; no production database, migrations, or security layer is claimed.
- 18 SpotBugs warnings remain disclosed; no claim of zero static-analysis findings is made.
- No screenshots were captured automatically; genuine command excerpts and committed raw logs are used instead.

Conclusion

The branch contains a working, tested, documented, container-verified Library API and a traceable eight-stage engineering history. The final state is reproducible with the Maven Wrapper and Dockerfile, while uncertainties and unresolved risks remain explicit.

GitHub branch
https://github.com/Yorgo756/genie-logiciel-reference-lab/tree/yorgo/reference-lab